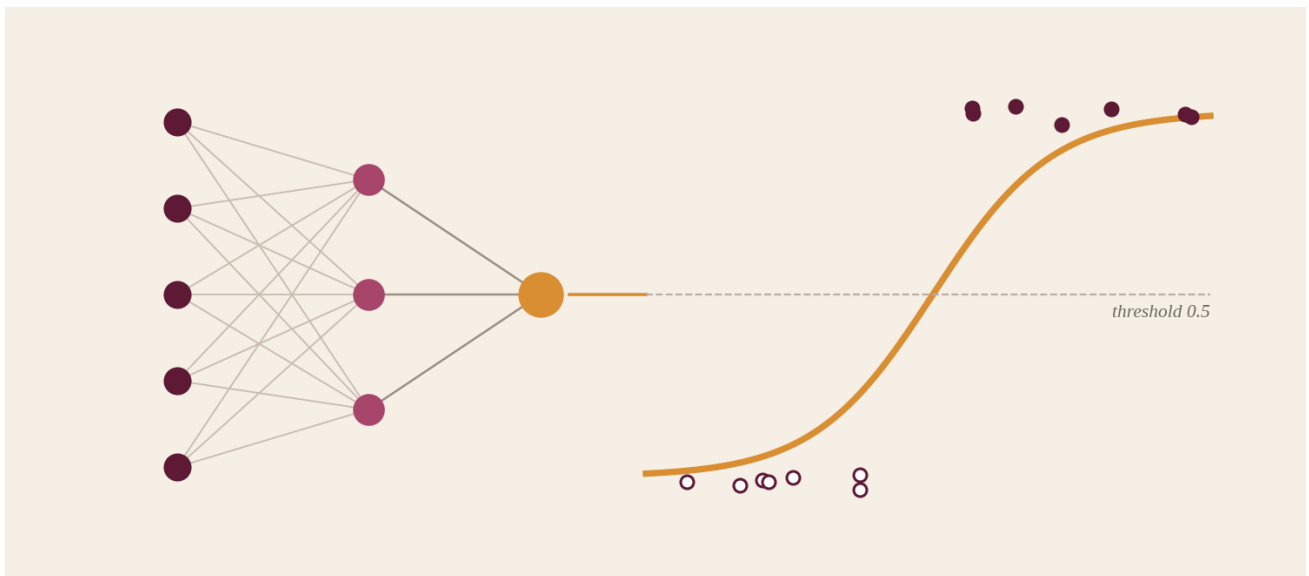
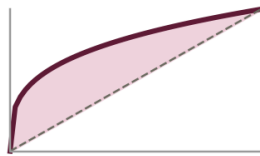


RISK ALERT CLASSIFIER

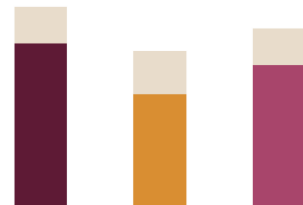
Part :- A Conceptual Understanding (Theory)



Confusion
Matrix



ROC - AUC



Precision · Recall · F1

Part A: Conceptual Understanding (Theory)

1 What is **Logistic Regression** and why is it suitable for classification?

Logistic Regression is a supervised algorithm that predicts the **probability** of a class. It passes a linear combination of features through the sigmoid function, so the output always lies between 0 and 1. A threshold (e.g. 0.5) then turns this probability into a class label, which makes it simple, fast and interpretable for Yes/No problems such as risk prediction.

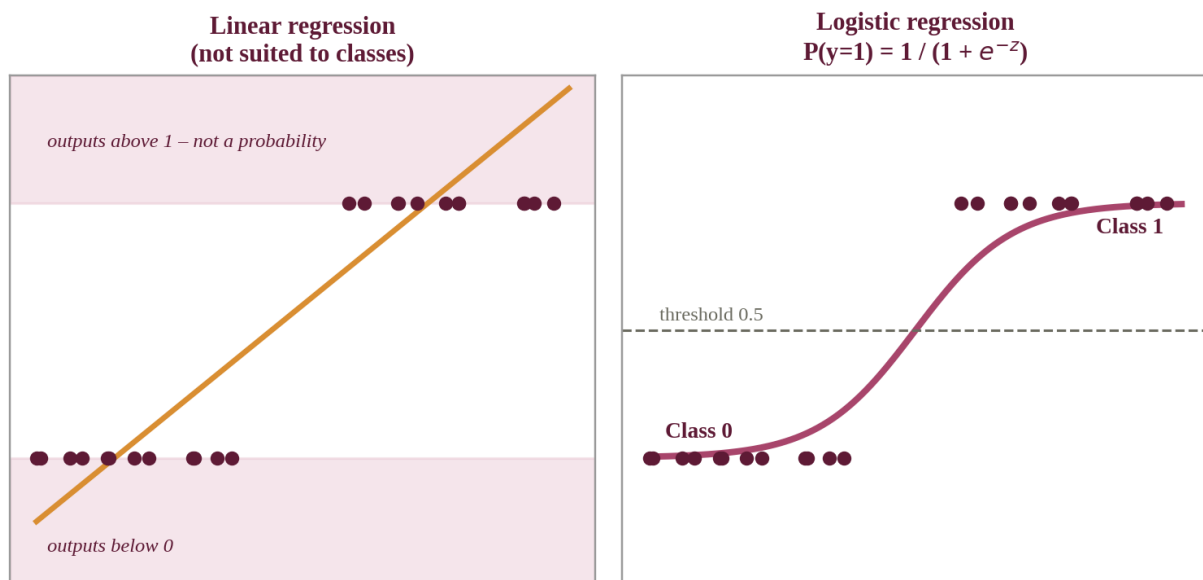


Figure 1. A straight line gives values outside 0–1; the S-shaped sigmoid gives valid probabilities.

2 Explain **classification performance metrics** and why accuracy alone is insufficient.

Classification metrics measure how well predictions match the true labels. Most are built from the confusion matrix (TP, FP, FN, TN): **Accuracy, Precision, Recall, F1-Score and AUC-ROC**. Accuracy only counts the share of correct predictions, so on imbalanced data a model that always predicts the majority class can score 95% while catching none of the important cases.

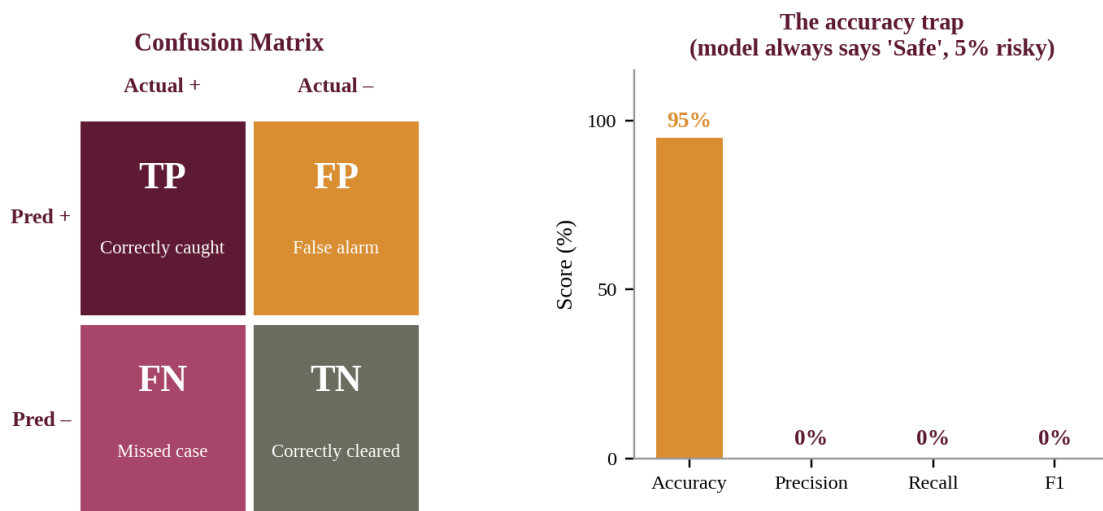


Figure 2. The confusion matrix, and how 95% accuracy can hide 0% recall.

3 Define **Type-I Error** and **Type-II Error** in the context of risk prediction.

Type-I Error (False Positive): a safe customer is wrongly predicted as risky, causing a false alarm and a lost good customer. **Type-II Error (False Negative):** a risky customer is wrongly predicted as safe, e.g. a loan is approved and later defaults. In risk prediction, Type-II errors are usually the costlier mistake.

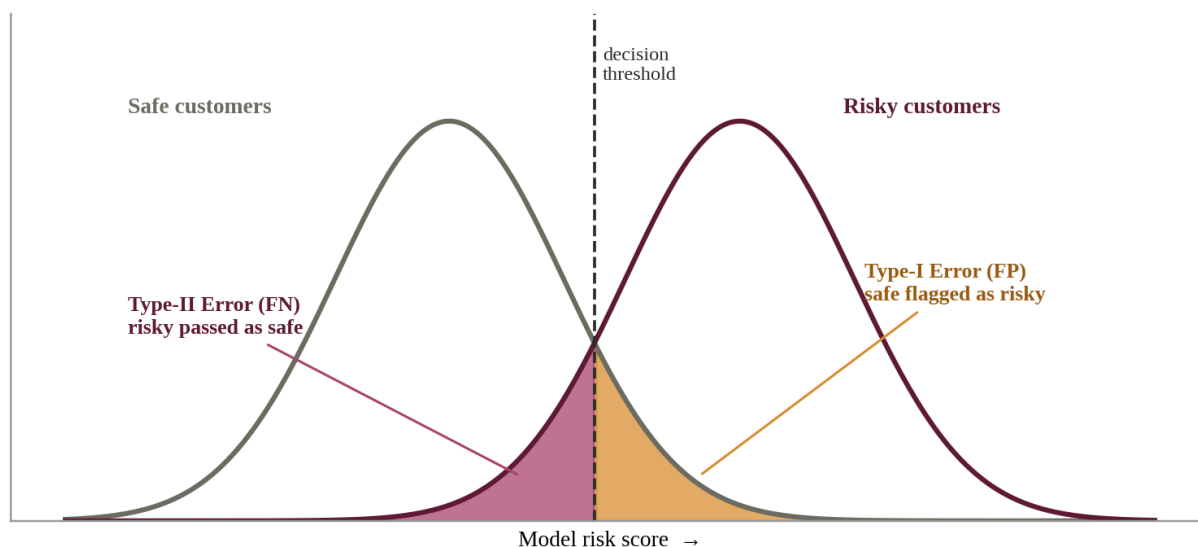


Figure 3. Overlapping score distributions: moving the threshold trades one error for the other.

4 Explain **Precision**, **Recall**, **F1-Score**, **TPR**, and **FPR**.

Precision is the share of predicted positives that are truly positive. **Recall (TPR)** is the share of actual positives that were caught. **F1-Score** is their harmonic mean, balancing the two. **FPR** is the share of actual negatives wrongly flagged as positive.

	Precision $TP / (TP + FP)$			Recall = TPR $TP / (TP + FN)$			FPR $FP / (FP + TN)$	
	Act +	Act -		Act +	Act -		Act +	Act -
Pred +	TP	FP	Pred +	TP	FP	Pred +	TP	FP
Pred -	FN	TN	Pred -	FN	TN	Pred -	FN	TN

dark = numerator, light pink = denominator. $F1-Score = 2 \times Precision \times Recall / (Precision + Recall)$

Figure 4. Each metric reads a different part of the confusion matrix.

5 What is **AUC-ROC** and how does it help in evaluating classifiers?

The **ROC curve** plots TPR against FPR at every possible threshold, and **AUC** is the area under it: 1.0 is perfect and 0.5 is random guessing. It helps because it judges a model across all thresholds at once, does not depend on

one chosen cut-off, and makes comparing classifiers easy. A curve hugging the top-left corner means a better model.

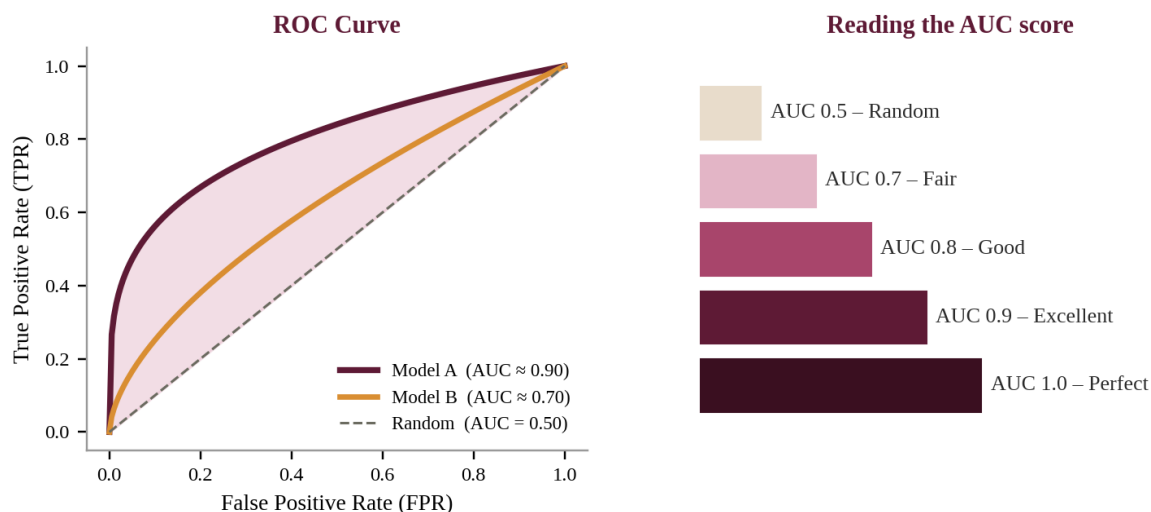


Figure 5. A higher curve and larger area mean better class separation.

6 Why does **imbalanced data** create problems in classification models?

When one class heavily outnumberes the other (e.g. 95% safe, 5% risky), the model is pulled toward the majority class because that minimises overall error. It learns little about the minority class, giving high accuracy but poor recall. Fixes include **resampling (SMOTE / undersampling)**, **class weights**, and evaluating with F1 or AUC instead of accuracy.

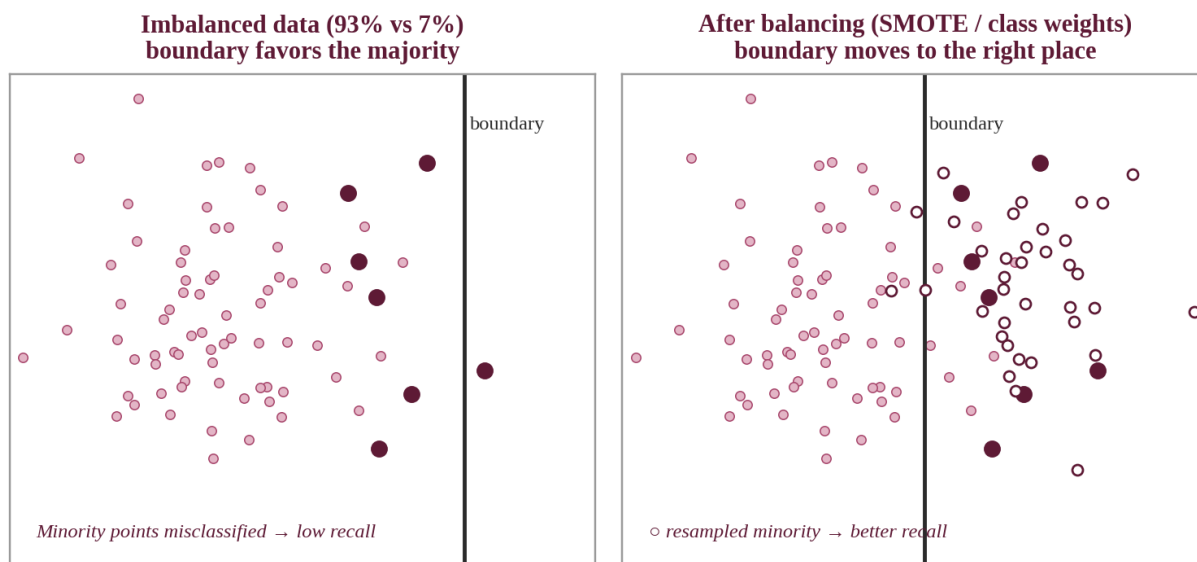


Figure 6. Balancing the classes moves the decision boundary so the minority class is recognised.